

packages/graph-view/src/utils/elk-layout.ts

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Overview

`packages/graph-view/src/utils/elk-layout.ts` now offloads ELK layout to a WebWorker when the node count exceeds a threshold, adds a lightweight grid fallback for worker failures, and improves error handling. The file also pulls in new performance helpers and worker-specific types.

Key changes

- **Imports:** added `countElkLayoutNodes` and `shouldRunElkInWorker` from `./performance`; added `ElkWorkerRequest/ElkWorkerResponse` types from `./elk-layout.worker`.
- **Worker plumbing:** introduced `ensureLayoutWorker`, `runElkLayoutInWorker`, and `runElkLayoutOnMainThread`. A `layoutWorkerPending` map tracks promises per request ID.
- **Fallback logic:** new `applyElkGridFallback` lays out nodes on a simple grid when ELK fails.
- **applyElkLayout refactor** (lines 303-328):
 - Counts nodes (`countElkLayoutNodes`) and chooses worker or main-thread execution via `shouldRunElkInWorker`.
 - On error, if `opts.strict` is true, re-throws; otherwise returns the grid fallback and appends a fatal issue.
 - Removed the old “empty children” error return.
- **Worker error handling:** `layoutWorker.onerror` now rejects all pending promises and clears the worker.

Impact

- **Performance:** Large graphs now run ELK in a worker, preventing UI blocking.
- **Reliability:** Grid fallback guarantees a non-empty layout even if ELK crashes or the worker fails.
- **Observability:** Fatal issues are still surfaced, and the worker’s error path logs a clear message.
- **Maintainability:** Centralized worker logic and helper imports reduce duplication.

Risks & follow-ups

- **Worker creation:** Verify that the worker URL resolves correctly in all bundlers; test in both dev and production builds.
- **Threshold tuning:** `shouldRunElkInWorker` may need adjustment for edge cases; monitor node-count distributions.
- **Fallback correctness:** Ensure the grid layout does not introduce visual artifacts; run visual regression tests.
- **Error propagation:** Confirm that non-strict mode still surfaces the fatal issue in logs and UI.